1. Introduction

- mathematical modeling
- model descriptions
- disciplined machine learning
- examples
- course goals and topics
- mathematical optimization
- convex optimization
- brief history of convex optimization
- nonlinear optimization

Mathematical modeling

build (mathematical) **models**

- that can
  - learn from data
  - make predictions or decisions based on the learned knowledge

- usually represented as functions that map some *inputs* (or *features*) to outputs (or *responses*, *observations*)

- characterized by a set of *parameters* that determine their behavior

Model descriptions

natural language descriptions define a model using math symbols and texts, $e.g.$,

> Given observations $a_1, \ldots, a_m \in \mathbf{R}^n$ and $y \in \mathbf{R}^m$, a linear regression model assumes $y_i \approx a_i^T x$ and chooses $x \in \mathbf{R}^n$ to minimize the sum of squared errors.

algorithmic descriptions define a model in terms of a procedure to compute the model's parameters, $e.g.$,

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Linear regression.

given $A = \begin{bmatrix} a_1 & \cdots & a_m \end{bmatrix}^T \in \mathbf{R}^{m \times n}$, $y \in \mathbf{R}^m$

return $x := (A^T A)^{-1} A^T y$

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inverse problem descriptions define a model in terms of the associated fitting problem, $e.g.$,

$$\text{minimize} \quad \sum_{i=1}^{m} \left(a_i^T x - y_i\right)^2$$

with variable $x \in \mathbf{R}^n$ and data $a_1, \ldots, a_m \in \mathbf{R}^n$ and $y \in \mathbf{R}^m$ (describes a linear regression model)

Disciplined machine learning

$$\begin{array}{ll} \text{minimize} & f_0(x) + \phi(x) \\ \text{subject to} & x \in C \end{array}$$

- $x \in \mathbf{R}^n$ is the model *parameter*
- $f_0 \colon \mathbf{R}^n \to \mathbf{R}$ is the *cost* or *loss function*
- $\phi \colon \mathbf{R}^n \to \mathbf{R}$ is the *regularization function* or *regularizer*
- $C \subseteq \mathbf{R}^n$ is the *constraint set*

- f_0 represents the primary objective of the model, such as fitting or prediction error
- ϕ and C represent prior information of required model properties

Least squares

$$\text{minimize} \quad \|Ax - b\|_2^2$$

- variable $x \in \mathbf{R}^n$
- data $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$
- a convex optimization problem
- analytical solution $x^\star = (A^T A)^{-1} A^T b$

- easy to recognize and a mature technology
- reliable and efficient algorithms and software

regularized least squares

$$\text{minimize} \quad \|Ax - b\|_2^2 + \lambda \|x\|_2^2$$

- $\|x\|_2^2$ called ridge or Tikhonov regularization
- analytical solution $x^\star = (A^T A + \lambda I)^{-1} A^T b$
- induce 'small' solutions $x^\star$

$$\text{minimize} \quad \|Ax - b\|_2^2 + \lambda \|x\|_1$$

- $\|x\|_1$ called lasso regularization
- no analytical solution, but is a convex optimization problem
- can be solved efficiently by numerical methods
- induce sparse solutions $x^\star$ (many components are zero)

(linear inequality) constrained least squares

$$\begin{array}{ll} \text{minimize} & \|Ax-b\|_2^2 \\ \text{subject to} & Cx \preceq d \end{array}$$

- a convex optimization problem; no analytical solution
- $A = I$ and $b = 0$, $\preceq$ takes equality $\Longrightarrow$ least norm problem

$$\begin{array}{ll} \text{minimize} & \|x\|_2^2 \\ \text{subject to} & Cx = d \end{array}$$

  finds the minimum (ℓ_2-) norm solution of the linear system $Cx = d$

- $C = -I$ and $d = 0$ $\Longrightarrow$ nonnegative least squares problem

$$\begin{array}{ll} \text{minimize} & \|Ax-b\|_2^2 \\ \text{subject to} & x \succeq 0 \end{array}$$

  finds the best nonnegative solution of the linear system $Ax = b$

Principal component analysis (PCA)

$$\begin{array}{ll} \text{minimize} & \|Z - A\|_F^2 \\ \text{subject to} & \mathbf{rank}\, Z \leq k \end{array}$$

- variable $Z \in \mathbf{R}^{m \times n}$
- data (matrix) $A \in \mathbf{R}^{m \times n}$, rank k ($\ll n$)
- $\|\cdot\|_F$ is the Frobenius norm, *i.e.*,

$$\|U\|_F = \left(\mathbf{tr}(U^T U)\right)^{1/2} = \left(\sum_{i=1}^{m}\sum_{j=1}^{n} U_{ij}^2\right)^{1/2}$$

- finds the best rank-k approximation of $A \in \mathbf{R}^{m \times n}$

factorized formulation of PCA

$$\text{minimize} \quad \|XY - A\|_F^2$$

- variables $X \in \mathbf{R}^{m\times k}$, $Y \in \mathbf{R}^{k\times n}$
- *not* a convex problem
- but has analytical solution

$$X^\star = U_k\Sigma_k^{1/2}, \qquad Y^\star = \Sigma_k^{1/2}V_k^T$$

from SVD $A = U\Sigma V^T$

- $U_k \in \mathbf{R}^{m\times k}$, $V_k \in \mathbf{R}^{n\times k}$ are the first k left and right singular vectors
- $\Sigma_k = \mathbf{diag}(\sigma_1, \ldots, \sigma_k) \in \mathbf{R}^{k\times k}$ is the diagonal matrix of the first k singular values

regularized PCA

- quadratically regularized PCA

$$\text{minimize} \quad \|XY - A\|_F^2 + \lambda(\|X\|_F^2 + \|Y\|_F^2)$$

  - not a convex problem, also has analytical solution via SVD
  - finds a low-rank approximation of A with small entries in X and Y

- sparse PCA

$$\text{minimize} \quad \|XY - A\|_F^2 + \lambda(\|X\|_{\text{sav}} + \|Y\|_{\text{sav}})$$

  - sum-absolute-value norm $\|U\|_{\text{sav}} = \sum_{i=1}^{m} \sum_{j=1}^{n} |U_{ij}|$
  - not a convex problem; good approximation method exists
  - finds a low-rank approximation of A with sparse entries in X and Y

constrained PCA: nonnegative matrix factorization

$$\begin{array}{ll} \text{minimize} & \|XY - A\|_F^2 \\ \text{subject to} & X_{ij} \geq 0, \quad i = 1, \ldots, m, \quad j = 1, \ldots, k \\ & Y_{ij} \geq 0, \quad i = 1, \ldots, k, \quad j = 1, \ldots, n \end{array}$$

- finds nonnegative low-rank approximation of A
- nonconvex problem; good approximation method exists

Course goals and topics

goals

- recognize/formulate disciplined inverse problems
- develop code scripts to solve inverse problems
- characterize solution quality

topics

- mathematical optimization
  - convex sets, functions, and optimization problems
  - sequential methods for nonlinear optimization
- disciplined modules
  - objective functions and regularizers
  - constraints, relaxation, and duality
- applications, $e.g.$, robust and latent factor models

Mathematical optimization

(mathematical) optimization problem

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \ldots, m \\ & h_i(x) = 0, \quad i = 1, \ldots, p, \end{array}$$

- $x = (x_1, \ldots, x_n) \in \mathbf{R}^n$: *optimization variables*
- $f_0 : \mathbf{R}^n \to \mathbf{R}$: *objective function*
- $f_i : \mathbf{R}^n \to \mathbf{R}$, $i = 1, \ldots, m$: *inequality constraint functions*
- $h_i : \mathbf{R}^n \to \mathbf{R}$, $i = 1, \ldots, p$: *equality constraint functions*

optimal point $x^\star$ has smallest value of f_0 among all vectors that satisfy the constraints

Solving optimization problems

general optimization problem

- very difficult to solve
- usually involve some compromise
  - very long computation time
  - only finds an approximate solution

exceptions: certain problem classes can be solved efficiently and reliably

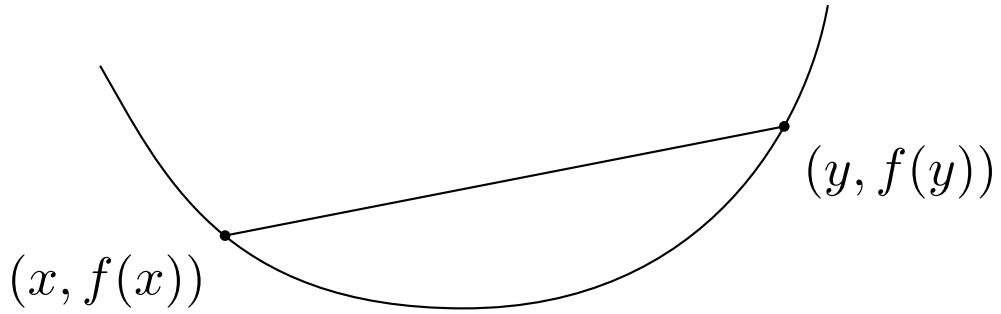
Convex optimization

convex optimization problem satisfies

- $f_i : \mathbf{R}^n \to \mathbf{R}$, $i = 0, \ldots, m$, are convex, *i.e.*,

$$f_i(\theta x + (1-\theta)y) \leq \theta f_i(x) + (1-\theta) f_i(y)$$

  for all $x, y \in \mathbf{R}^n$ and $\theta \in [0, 1]$



- $h_i : \mathbf{R}^n \to \mathbf{R}$, $i = 1, \ldots, p$, are affine

examples: least squares, least norm problems, . . .

solving convex optimization problems

- (usually) no analytical solution
- reliable and efficient numerical algorithms
- a technology; we could now say

  If a problem can be shown/formulated convex, then it is solved.

using convex optimization

- often difficult to recognize
- many tricks for transforming problems into convex form
- surprisingly many problems can be solved via convex optimization (however less known)

Brief history of convex optimization

theory (convex analysis): approximately 1900–1970

algorithms (convex optimization)

- 1947: simplex algorithm for linear programming (Dantzig)
- 1960s: early interior-point methods (Fiacco & McCormick, Dikin, . . .)
- 1970s: ellipsoid method and other subgradient methods
- 1980s: polynomial-time interior-point methods for linear programming (Karmarkar 1984)
- late 1980s–now: polynomial-time interior-point methods for nonlinear convex optimization (Nesterov & Nemirovski 1994)

applications

- before 1990: mostly in operations research; few in engineering

- since 1990:
  - many new applications in engineering (control, signal processing, communications, circuit design, . . .)
  - new problem classes
    * semidefinite and second-order cone programming; robust optimization
    * exponential and power cone programming (many applications in machine learning)

disciplined convex programming: 2000–now (Boyd & Vandenberghe, Grant, Diamond, . . .)

- human-readable symbolic modeling languages for convex optimization
- since 2010: extensions to many nonconvex problem classes (geometric, quasiconvex, convex-concave, . . .)
- close to the idea

  Say what you want, not how to get it.

Nonlinear optimization

general nonconvex optimization involve compromises

local optimization (traditional nonlinear optimization)

- find a point that minimizes f_0 among feasible points *near* it
- fast, can handle large problems
- require initial guess that influences the solution quality
- no information about distance to (global) optimum

global optimization

- find the (global) solution
- worst-case complexity grows exponentially with problem size

these methods are often based on solving convex subproblems